

APPROACH TO HEADACHES IN THE CHILDREN'S EMERGENCY

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1. HISTORY OF CLINICAL PRESENTATION

(Refer to **ANNEX A** for diagram on general approach.)

- Goals: establish headache pattern, presence of red flags, determine if primary vs. secondary headaches
- History taking is key to making the diagnosis and should be provided by the patient first before parent subsequently adds on after
- Patients with red flags should be referred to a pediatric neurologist

1.1 PEARLS

- Pre-school children rarely complaint of headache – it should be taken seriously
- Patients may have more than 1 type of primary headache
- Intervals between primary headache episodes should be pain & symptom free
- Patients presenting with neurological symptoms should be assessed for organic causes before the diagnosis of migraine with aura is formally established.
 - Evaluation by pediatric neurologist and/or
 - Evaluation with neuroimaging

1.2 HEADACHE PATTERNS

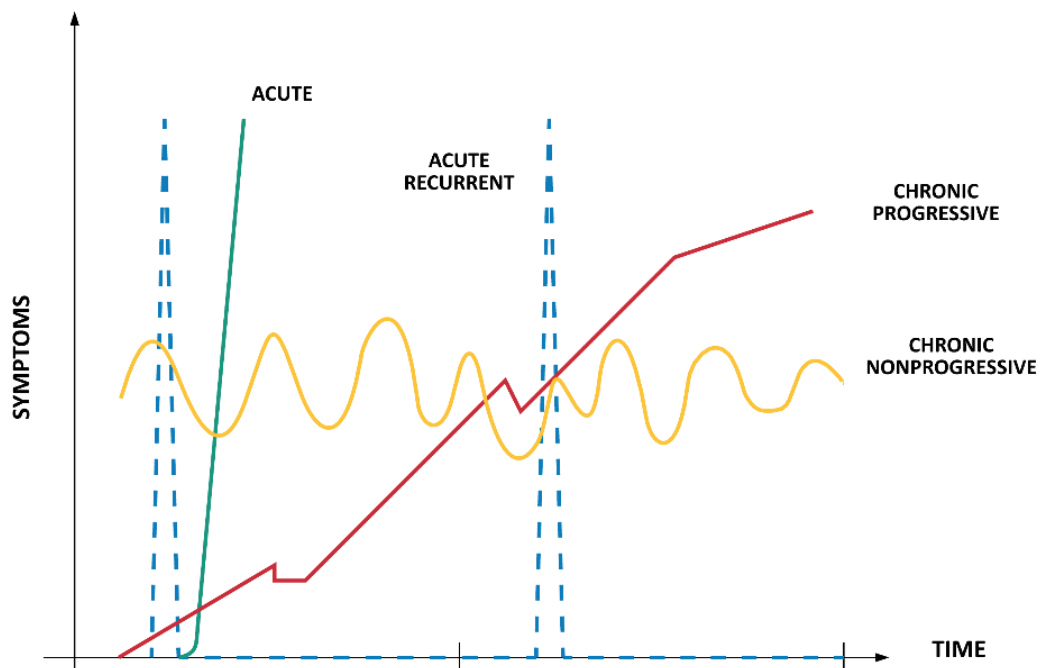


Figure 1 - Headache Patterns. (Adapted from Rothner AD. The evaluation of headaches in children & adolescents. *Semin Pediatr Neurol.* 1995;2[2]:109-118)

1.3 RED FLAGS

1.3.1 HEADACHE CHARACTERISTICS

- Presentation in < 3 years
- First severe or worst headache in patient's life
- Wakes the child up from sleep or occurs consistently upon awakening
- Associated with persistent nausea/vomiting, altered mental status
- Headache worse in recumbent position or exacerbated with cough, micturition, or defecation
- New persistent or progressive worsening headache
- Change in character from usual established headache pattern

1.3.2 FUNCTION

- Focal neurology/ataxia
- Academic decline
- Personality/behavioral change

2. CLASSIFICATION OF PRIMARY HEADACHES

ICHD-3 CRITERIA FOR HEADACHE DISORDERS: MIGRAINE		
CRITERIA FOR PEDIATRIC MIGRAINE WITHOUT AURA		
1	≥ 5 episodic attacks	
2	Headaches lasting 2 – 72 hours (untreated or unsuccessfully treated)	If patient falls asleep during headache and wakes up without it, duration of attack is reckoned until the time of awakening.
3	≥ 2 of the following characteristics	▫ Unilateral location (can be bilateral in pediatric population) ▫ Pulsating quality ▫ Moderate or severe pain intensity ▫ Aggravated by or causing avoidance of routine physical activity (e.g., walking or climbing stairs)
4	≥ 1 associated symptom	▫ Nausea or vomiting ▫ Photophobia and phonophobia
5	Not explained by any other ICHD-3 diagnosis	
CRITERIA FOR MIGRAINE WITH AURA		
1	Headache fulfilling migraine criteria	
2	≥ 1 reversible symptom	▫ Visual * ▫ Sensory ▫ Motor ▫ Retinal ▫ Speech and/or language ▫ Brainstem
3	≥ 3 characteristics	▫ ≥ 1 aura symptoms spread over ≥ 5 mins ▫ ≥ 2 aura symptoms occurring in succession ▫ Each individual aura symptom lasts 5 – 60 mins ▫ ≥ 1 symptom is unilateral ▫ ≥ 1 symptom is positive (scintillations, pins-and-needles) ▫ Aura is accompanied or followed within 60 mins by headache

ICHD-3 CRITERIA FOR: TENSION-TYPE HEADACHE (TTH)

Headache has ≥ 2 characteristics:		▫ Bilateral location ▫ Pressing or tightening (non-pulsating) quality ▫ Mild or moderate intensity ▫ Not aggravated by routine physical activity			
	TYPE	FREQUENCY	DURATION	NAUSEA	PHOTO/PHONOPHOBIA
1	Infrequent episodic*	≤ 10 episodes ≤ 1 day per month on average (< 12 days/year)	30 min – 7 days	No	≤ 1 (but not both)
2	Frequent episodic	≤ 10 episodes occurring on 1 – 14 days per month on average for > 3 months	30 min – 7 days	No	≤ 1 (but not both)
3	Chronic	≥ 15 days per month for > 3 months (≥ 180 days/year)	Hours – days or unremitting	Mild	≤ 1 (but not both)

*Occurs in almost the entire population with little impact on the individual & patients typically do not require medical management

ICHD-3 CRITERIA FOR: TRIGEMINAL AUTONOMIC CEPHALALGIA (TAC)

Unilateral orbital, supra-orbital and/or temporal headache with ≥ 1 characteristic:		▫ At least one ipsilateral autonomic sign/symptom: ▪ Conjunctival injection and/or lacrimation ▪ Nasal congestion and/or rhinorrhea ▪ Eyelid edema ▪ Forehead & facial sweating ▪ Miosis and/or ptosis ▫ A sense of restlessness or agitation			
	TAC TYPE	NO. OF ATTACKS	SEVERITY	FREQUENCY ^e	DURATION ^f
1	Cluster ^a	≥ 5	Very severe	1 every other day – 8 per day	15 – 180 mins (untreated)
2	Paroxysmal hemicrania ^{b c}	≥ 20	Severe	≥ 5 per day	2 – 30 mins
3	Short-lasting, unilateral, neuralgiform headache attacks + <u>both</u> conjunctival injection & tearing (SUNCT) + <u>either</u> conjunctival injection or tearing (SUNA)	≥ 20	Moderate to severe Unilateral orbital, supra-orbital, temporal and/or other trigeminal distribution	≥ 1 per day	1 – 600 sec Single or multiple stabs Sawtooth pattern
4	Hemicrania continua ^{c d}	Persistent	Exacerbation of $\geq$ moderate intensity	NA	≥ 3 months

^a Rare in patients < 10 years and uncommon in older patients; Male to Female = 4:1

^b Onset usually in adulthood, 20 – 40 years old; childhood cases reported

^c Absolute response to indomethacin.

^d Continuous pain with exacerbations.

^e In increasing attack frequency (cluster, paroxysmal hemicrania, SUN)

^f In decreasing attack duration (cluster, paroxysmal hemicrania, SUN)

3. ESSENTIAL INVESTIGATIONS

- Neuroimaging is required for evaluation of intracranial causes of secondary headaches
- CT is often the primary modality of choice in an acute or emergent setting for detection of tumors, hydrocephalus, intracranial hemorrhages, vascular abnormalities

3.1 INDICATIONS FOR NEUROIMAGING

- Focal neurology on examination
- Signs of raised intracranial pressure:
 - Papilledema
 - Cushing reflex (bradycardia, hypertension)
- Thunderclap headache (especially if associated with neck pain or stiffness)
- Red flag symptoms:
 - Recurrent morning headaches associated with nausea ± vomiting
 - Headaches that result in nighttime awakening
 - Acute increase in severity on a background of chronic progressive headache

3.2 LIMITATIONS OF CT BRAIN

- Poorly images the posterior fossa because of artifacts produced from the surrounding thick bone
- Involves radiation exposure
- Requires contrast for vascular pathology
- Early infarcts not visualized

4. ACUTE MANAGEMENT OF HEADACHES

4.1 MIGRAINES

4.1.1 PRINCIPLES OF TREATMENT

- Acute migraine treatments are more likely to be effective when used earlier (within 1H of headache onset).mg
- There is no evidence that supports the use of opioids in the treatment of children with migraine.

4.1.2 ANALGESICS

4.1.2.1 ORAL

- PO naproxen sodium (Synflex®) [1st line for adolescents] (max. 1.1 g/day)
 - ≥ 18 years: PO naproxen sodium 275 mg TDS or 550 mg BD
 - 2 – 17 years: PO naproxen sodium 5.5 – 8.25 mg/kg BD

Note: Naproxen sodium 275mg (Synflex®) = Naproxen 250mg

- PO ibuprofen 10 mg/kg (400 mg) 6 – 8H
- PO paracetamol 10 – 15 mg/kg (1 g) 6H

4.1.2.2 INTRAVENOUS

For patients who are unable to tolerate oral medication

- IV paracetamol (dose & frequency as per PO formulation; infusion over 15 – 30 mins)
- IV ketorolac
 - Serve in addition to IV paracetamol with gastric protection (IV omeprazole)
 - Total duration of ketorolac should not exceed 5 days
 - Can be served at intervals of 6 – 8H
 - Dose:
 - ≥ 2 to ≤ 16 years: IV ketorolac 0.5 mg/kg (max. 30mg/dose) once
 - ≥ 17 years:
 - < 50kg: IV ketorolac 15 mg once (max. daily dose = 60 mg)
 - ≥ 50kg: IV ketorolac 30 mg once (max. daily dose = 120 mg)

4.1.3 TRIPTANS

Note: Causes vasoconstriction and could aggravate or worsen symptoms that may be caused by vasospasm. Contraindicated in patients with a history of ischemic vascular disease.

4.1.3.1 GUIDELINES FOR USE

- Consider in patients ≥ 12 years with an established diagnosis of migraine
- Consider in patients ≥ 17 years with suspected migraine with normal neuroimaging
- Do not use within 24H of another triptan, or with other ergots
- Contraindications:
 - History of cardiac conduction disorders
 - Uncontrolled hypertension
 - Basilar or hemiplegic migraine or migraine with prolonged aura
- Adverse effects (last < 30 mins in general)
 - Paresthesia
 - Sleepiness, numbness
 - Warm/hot sensation
 - Weakness/myalgia/fatigue
 - Tightness in chest or throat

4.1.3.2 SUMATRIPTAN

- 1st dose ASAP after onset of headache
- 2nd dose can be re-dosed after 2H if
 - 6 – 17 years: if headache recurs
 - ≥ 18 years: if headache does not respond or if headache recurs
- Age-based dosing
 - 6 – 9 years: 25 mg per dose
 - 10 – 11 years: 50 mg per dose
 - 12 – 17 years: 50 – 100 mg per dose
 - ≥ 18 years: Initial single dose of 25mg, 50mg or 100mg
(max. 200 mg/day)

4.1.4 COMBINATION THERAPY

- For acute migraine attacks refractory to monotherapy
- PO naproxen 5 mg/kg taken together with a triptan

4.1.5 ANTI-EMETICS

- PO promethazine 0.2 – 0.5 mg/kg (10 – 25 mg) 6 – 8H (> 2-years-old)
- PO ondansetron 0.1 – 0.2 mg/kg (8 mg) 8H

4.2 TENSION-TYPE HEADACHES (TTH)

- PO medication (naproxen, ibuprofen, paracetamol)

4.3 TRIGEMINAL AUTONOMIC CEPHALAGIA (TAC)

- An uncommon and difficult diagnosis to make in the CE setting; to consult Pediatric Neurologist if diagnosis is suspected and acute treatment is required.
- PO NSAIDs (ibuprofen or naproxen)
- Oxygen therapy
 - FiO₂ 100% via NRM @ 12 L/min of O₂ x 15 minutes continuously
 - Pain may stop after 5 minutes
- PO sumatriptan (dose as per migraine dose)
- Paroxysmal hemicrania and hemicrania continua
 - PO indomethacin 0.5 – 1 mg/kg (25 – 50 mg) 12H (max. 4 mg/kg/day or 200 mg/day)

5. LIFESTYLE MODIFICATION ADVICE FOR MIGRAINE

5.1 TRIGGER AVOIDANCE

- Stress
- Sleep disturbance / deprivation
- Caffeine
- Weather change
- Sunlight

5.2 DIET & HYDRATION

- Caffeine withdrawal & hunger/fasting can trigger migraines
- Avoid caffeine-containing products
- Avoid skipping meals/fasting

5.3 EXERCISE

- Intensity: break into sweat and slightly out of breath
- 30 mins, 3 – 4 times per week

5.4 SLEEP

5.4.1 AVERAGE SLEEP DURATION

PRIMARY SCHOOL AGE

Age (years)	Hours
7	10.5
8	10.25
9	10
10	9.75
11	9.5
12	9.25

SECONDARY SCHOOL AGE & ABOVE

Age (years)	Hours
13	9.25
14	9
15	8.75
16	8.5
17	8.25
18	8.25

5.4.2 SLEEP HYGIENE¹

- i. Have a set bedtime and bedtime routine.
- ii. Bedtime and wake-up time should be approximately the same time on school nights and non-school nights. There should not be more than 1h difference from 1 day to another.
- iii. Shared quiet time 1h before bedtime. Avoid high-energy activities (rough play) or stimulating activities (watching TV or computer games).
- iv. Do not go to bed hungry – a light snack would suffice. A heavy meal within 2h of bedtime can interfere with sleep.
- v. Avoid products containing caffeine.
- vi. Regular exercise and as much outdoor activity during the day.
- vii. Keep bedroom quiet and dark.
- viii. Keep temperature comfortable.
- ix. Do not use bedroom for punishment or time-out.
- x. Avoid use of electronic devices within 1h of bedtime. Keep devices out of bedroom.

¹ Owens J, Mindell J. Sleep Hygiene: Healthy Sleep Habits for Children and Adolescents. In: A Clinical Guide to Pediatric Sleep, 2nd ed, Lippincott, Williams & Wilkins, Philadelphia 2010.

6. HEADACHE ATTRIBUTED TO HEAD TRAUMA

6.1 CRITERIA

- Traumatic injury to the head
- Headache onset within 7 days of:
 - Injury to the head
 - Regaining of consciousness after injury to the head
 - Discontinuation of medications that blunt senses to headache
- Headaches:
 - Resolved within 3 months
 - Not yet resolved, but 3 months have not yet passed

6.2 POST-CONCUSSION SYNDROME

6.2.1 SYMPTOMS

- Headache
 - Migraine-like, tension-type-like
 - Other syndromes: TMJ injury, craniofacial pain
- Dizziness (benign paroxysmal positional vertigo, labyrinthine concussion)
- Sleep disturbance (insomnia, sleep-wake disorders, excessive daytime sleepiness)
- Psychological and cognitive symptoms

6.2.2 TREATMENT

6.2.2.1 HEADACHES

- Monotherapy (PO ibuprofen, PO paracetamol)
- Combination analgesia
 - PO Anarex (paracetamol 450 mg + orphenadrine 35 mg) (max. 2 tab) 8H
 - Orphenadrine dosing: 1 – 2 mg/kg

6.2.2.2 DIZZINESS

- PO cinnarizine 0.3 – 0.6 mg/kg (15 – 30 mg) 8H
- PO betahistine 0.15 – 0.3 mg/kg (8 – 16 mg) 8H

6.2.2.3 NAUSEA

- PO promethazine 0.2 – 0.5 mg/kg (10 – 25 mg) 6 – 8H (> 2-years-old)
- PO ondansetron 0.1 – 0.2 mg/kg (8 mg) 8H

7. DISPOSITION

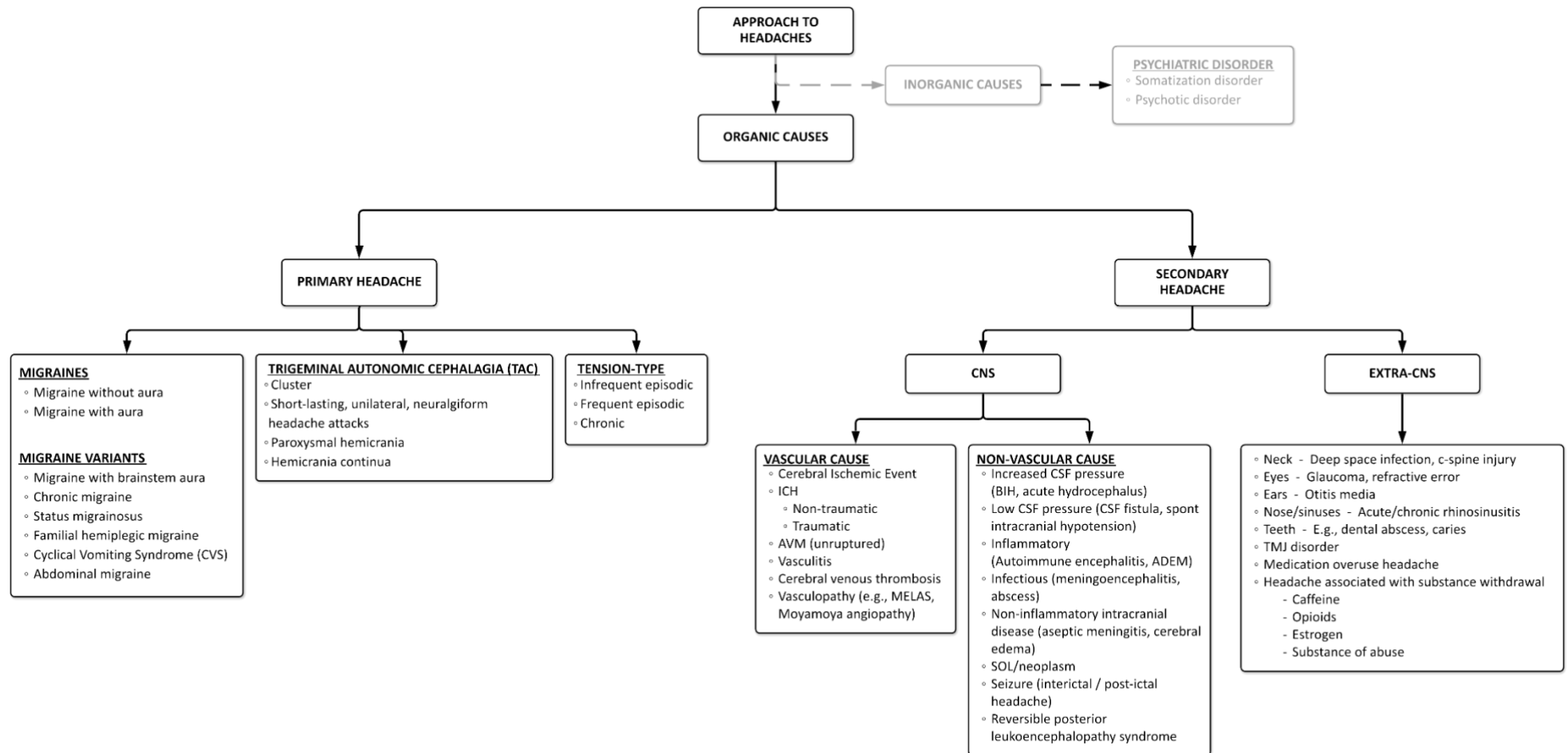
7.1 DISCHARGE

- Red flag symptoms & return advice
- Headache diary (ANNEX D)
- TCU Pediatric Neurology Outpatient Clinic x 1 week
 - For migraine with auras (neurologic symptoms)
- TCU General Pediatric Outpatient Clinic x 2 – 4 weeks
 - For tension headaches, migraines without aura

7.2 ADMISSION

- Secondary causes of headache suspected
- Consider CT brain before admission if indicated
- If GCS $\leq$ 14, to consider PICU/HD admission for frequent conscious level examination

ANNEX A GENERAL APPROACH TO HEADACHE (FLOWCHART)



ANNEX B CLINICAL FEATURES OF MIGRAINE WITH AURAS

ICHD-3 CRITERIA FOR HEADACHE DISORDERS: MIGRAINE		
CRITERIA FOR MIGRAINE WITH AURA		
1	Headache fulfilling migraine criteria	
2	≥ 1 reversible symptom	▫ Visual ▫ Sensory ▫ Speech and/or language ▫ Motor ▫ Retinal ▫ Brainstem
3	≥ 3 characteristics	▫ ≥ 1 aura symptoms spread over ≥ 5 mins ▫ ≥ 2 aura symptoms occurring in succession ▫ Each individual aura symptom lasts 5 – 60 mins ▫ ≥ 1 symptom is unilateral ▫ ≥ 1 symptom is positive (scintillations, pins-and-needles) ▫ Aura is accompanied or followed within 60 mins by headache

B1 PHASES OF MIGRAINE

- Phases of migraine (with aura) = prodromal → aura → headache → postdrome
 - Prodromal from a few hours to 2 days prior to aura
 - Postdrome up to 48h

B2 TYPICAL MIGRAINE AURAS

TYPE OF AURA*	POSITIVE	→	NEGATIVE [^]
VISUAL	▫ Photopsia ▫ Scintillations ▫ Fortification spectra ▫ Visual hallucinations/distortions		▫ Scotoma
SENSORY⁺	▫ Tingling		▫ Paresthesia (numbness)
SPEECH			▫ Dysarthria ▫ Aphasia
MOTOR[#]			▫ Weakness

* in order of most common aura

[^] typical progression of aura from positive to negative symptoms

⁺ progression typically cheiro-oral (hand to perioral)

[#] classified as hemiplegic migraine

- Each aura progresses over 5 minutes and lasts for less than 1 hour (motor auras can last up to 72h)
- Aura progresses from one type to another (e.g., visual → sensory)

ANNEX C MIGRAINE VARIANTS

	MIGRAINES WITH (OTHER) AURAS	
BASILAR MIGRAINE	- No motor/retinal symptoms - At least 2 brainstem symptoms: - Dysarthria - Hyperacusis - Vertigo - Diplopia - Tinnitus - Ataxia - Decreased consciousness (GCS $\leq$ 13)	- Fulfils criteria for migraine with aura - Brainstem symptoms completely reversible - Onset usually 7 – 20 years
HEMIPLEGIC MIGRAINE (MOTOR AURA)	- With both criteria: - Fully reversible motor weakness - Fully reversible visual, sensory and/or speech/language symptom	- Fulfils criteria for migraine with aura - Motor symptoms can last up to 72 hours - Ask for a family history of hemiplegic migraine
	COMPLICATION	
STATUS MIGRAINOSUS	- Patient with migraine with/out auras - Unremitting for > 72 h - Pain and/or associated symptoms are debilitating	- Fulfils criteria for migraines with or without aura except for duration & severity - Remissions up to 12h due to medications and sleep are accepted
	EPISODIC SYNDROMES ASSOCIATED WITH MIGRAINE	
RECURRENT GI DISTURBANCES	$\geq$ 5 episodes $\geq$ 1 of the following symptoms: - Abdominal pain - Abdominal discomfort - Nausea - Vomiting - Normal GI examination & evaluation - Not attributed to any other disorder	
CYCLICAL VOMITING SYNDROME	$\geq$ 5 episodes of intense nausea & vomiting - Stereotypical in the individual patient & recurring with predictable periodicity - Nausea & vomiting $\geq$ 4 times per hour - Lasts $\geq$ 1h up to 10 days - Attacks $\geq$ 1 week apart - Symptom free intervals	History and examination do not show signs of GI disease
ABDOMINAL MIGRAINE	$\geq$ 5 attacks $\geq$ 2 characteristics - Midline, periumbilical or poorly localized - Dull or “just sore” quality - Moderate or severe intensity $\geq$ 2 signs/symptoms - Anorexia - Nausea - Vomiting - Pallor - Attacks last 2 – 72h (untreated or treated unsuccessfully) - Pain-free intervals	- Nausea, vomiting and vasomotor symptoms - Headache does not occur during these episodes
BENIGN PAROXYSMAL VERTIGO	$\geq$ 5 attacks - Vertigo (no warning), most severe at onset, resolves within minutes to hours with no LOC $\geq$ 1 of 5 signs/symptoms: - Nystagmus - Ataxia - Vomiting - Pallor - Fearfulness - Normal neurological examination and audiometric and vestibular functions between attacks	

ANNEX D HEADACHE DIARY



PATIENT'S NAME: _____

[illegible]

ANNEX E SUMMARY OF MEDICATIONS

DRUG	ROUTE	DOSE		
NSAIDs				
Ibuprofen <i>Syrup = 20 mg/mL</i> <i>1 tablet = 200 mg</i>	PO	10 mg/kg (400 mg) 6 – 8H		
Synflex (Naproxen sodium) <i>1 tab = 275 mg</i>	PO	2 – 17 years	5.5 – 8.25 mg/kg 12H (max. 1.1 g/day)	
		≥ 18 years	275 mg 8H or 550mg 12H	
Indomethacin	PO	2 – 17 years	0.5 – 1 mg/kg (25 – 50 mg) 12H (max. 4 mg/kg/day or 200 mg/day)	
Ketorolac <i>Total duration ≤ 5 days</i> <i>Intervals of 6 – 8H</i>	IV	2 – 16 years	0.5 mg/kg (max. 30mg/dose) once	
		≥ 17 years	< 50 kg	15 mg once (max. 60 mg/day)
			≥ 50 kg	30 mg once (max. 120 mg/day)
PARACETAMOL				
Paracetamol <i>Suspension = 24 mg/mL</i> <i>1 tablet = 500 mg</i>	PO	10 – 15 mg/kg (1 g/dose) 6H		
	IV	10 – 15 mg/kg (1 g/dose), infuse over 15 – 30 minutes, 6H		
Anarex <i>1 tab = Paracetamol 450mg</i> <i>+ Orphenadrine 35mg</i>	PO	2 tablets (max.) 8H - Paracetamol 10 – 15 mg/kg per dose - Orphenadrine 1 – 2 mg/kg per dose		
TRIPTANS				
Sumatriptan (max. 200 mg/day)	PO	6 – 9 years	25 mg/dose	Re-dose after 2H if headache recurs
		10 – 11 years	50 mg/dose	
		12 – 17 years	100/dose	
			25mg, 50mg or 100mg (Initial single dose)	
		≥ 18 years	Re-dose after 2H if no response or if headache recurs	
DIZZINESS				
Cinnarizine	PO	0.3 – 0.6 mg/kg (15 – 30 mg) 8H		
Betahistine	PO	0.15 – 0.3 mg/kg (8 – 16 mg) 8H		
NAUSEA/VOMITING				
Promethazine	PO	> 2 years	0.2 – 0.5 mg/kg (10 – 25 mg) 6 – 8H	
Ondansetron	PO/SL	> 6 months	0.1 – 0.2 mg/kg (8 mg) 8H	
	IV		0.1 – 0.2 mg/kg (8 mg) 8H	

8. REFERENCES

- 1) Headache Classification Committee of the International Headache Society (IHS). (2018). The International Classification of Headache Disorders, 3rd edition. Cephalagia.
- 2) Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology and the American Headache Society. (2019). *Practice guideline update summary: Acute treatment of migraine in children and adolescents*. American Academy of Neurology.
- 3) Thomas Koch, MD, FAAP, FAAN , Christopher B Oakley, MD. (1 June, 2018). *Pediatric migraine: Diagnostic criteria and treatment*. Retrieved 17 April, 2021, from Contemporary Pediatrics:
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